

Tarea 8 (Ejercicio 3)

♥ Matrices y determinantes. ♥

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Ejercicio 3. (100 puntos)

Determina el conjunto solución del siguiente sistema:

$$\begin{cases} x_1 + x_2 + 3x_3 - 2x_4 - x_5 = 1 \\ 5x_1 - 2x_2 + 3x_3 + 7x_4 + 8x_5 = 3 \\ -3x_1 - x_2 + 2x_3 + 7x_4 + 5x_5 = 2 \\ 5x_1 + 3x_2 + x_3 - 2x_4 + x_5 = 3 \end{cases}$$

Trabajaremos con la matriz ampliada:

$$\left(\begin{array}{ccccc|c} 1 & 1 & 3 & -2 & -1 & 1 \\ 5 & -2 & 3 & 7 & 8 & 3 \\ -3 & -1 & 2 & 7 & 5 & 2 \\ 5 & 3 & 1 & -2 & 1 & 3 \end{array} \right)$$

$$R_2 - 5R_1, R_3 + 3R_1, R_4 - 5R_1$$

$$\left(\begin{array}{ccccc|c} 1 & 1 & 3 & -2 & -1 & 1 \\ 0 & -7 & -12 & 17 & 13 & -2 \\ 0 & 2 & 11 & 1 & 2 & 5 \\ 0 & -2 & -14 & 8 & 6 & -2 \end{array} \right)$$

$$R_2 + 4R_3$$

$$\left(\begin{array}{ccccc|c} 1 & 1 & 3 & -2 & -1 & 1 \\ 0 & 1 & 32 & 21 & 21 & 18 \\ 0 & 2 & 11 & 1 & 2 & 5 \\ 0 & -2 & -14 & 8 & 6 & -2 \end{array} \right)$$

$$R_3 - 2R_1, R_4 + 2R_1$$

$$\left(\begin{array}{ccccc|c} 1 & 1 & 3 & -2 & -1 & 1 \\ 0 & 1 & 32 & 21 & 21 & 18 \\ 0 & 0 & 5 & 5 & 4 & 3 \\ 0 & 0 & -8 & 4 & 4 & 0 \end{array} \right)$$

$$R_1 - R_2$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & -29 & -23 & -22 & -17 \\ 0 & 1 & 32 & 21 & 21 & 18 \\ 0 & 0 & 5 & 5 & 4 & 3 \\ 0 & 0 & -8 & 4 & 4 & 0 \end{array}\right)$$

$$R_1 - 3R_4$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & -5 & -35 & -34 & -17 \\ 0 & 1 & 32 & 21 & 21 & 18 \\ 0 & 0 & 5 & 5 & 4 & 3 \\ 0 & 0 & -8 & 4 & 4 & 0 \end{array}\right)$$

$$R_1 + R_3$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & -30 & -30 & -14 \\ 0 & 1 & 32 & 21 & 21 & 18 \\ 0 & 0 & 5 & 5 & 4 & 3 \\ 0 & 0 & -8 & 4 & 4 & 0 \end{array}\right)$$

$$R_2 + 4R_4$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & -30 & -30 & -14 \\ 0 & 1 & 0 & 37 & 37 & 18 \\ 0 & 0 & 5 & 5 & 4 & 3 \\ 0 & 0 & -8 & 4 & 4 & 0 \end{array}\right)$$

$$5R_4 + 8R_3$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & -30 & -30 & -14 \\ 0 & 1 & 0 & 37 & 37 & 18 \\ 0 & 0 & 5 & 5 & 4 & 3 \\ 0 & 0 & 0 & 60 & 52 & 144 \end{array}\right)$$

$$\left(\frac{1}{2}\right)R_4$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & -30 & -30 & -14 \\ 0 & 1 & 0 & 37 & 37 & 18 \\ 0 & 0 & 5 & 5 & 4 & 3 \\ 0 & 0 & 0 & 30 & 26 & 72 \end{array}\right)$$

$$R_1 + R_4, \quad R_2 - R_4, \quad 6R_3 - R_4$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -4 & 58 \\ 0 & 1 & 0 & 7 & 11 & -54 \\ 0 & 0 & 30 & 0 & -2 & -54 \\ 0 & 0 & 0 & 30 & 26 & 72 \end{array} \right)$$

$$30R_2 - 7R_4$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -4 & 58 \\ 0 & 30 & 0 & 0 & 148 & -2154 \\ 0 & 0 & 30 & 0 & -2 & -54 \\ 0 & 0 & 0 & 30 & 26 & 72 \end{array} \right)$$

$$\frac{1}{30}R_2, \quad \frac{1}{30}R_3, \quad \frac{1}{30}R_4$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -4 & 58 \\ 0 & 1 & 0 & 0 & 148/30 & -2154/30 \\ 0 & 0 & 1 & 0 & -2/30 & -54/30 \\ 0 & 0 & 0 & 1 & 26/30 & 72/30 \end{array} \right)$$

Entonces tenemos las siguientes ecuaciones:

$$\begin{cases} x_1 - 4x_5 = 58 \\ x_2 + \frac{148}{30}x_5 = -\frac{2154}{30} \\ x_3 - \frac{2}{30}x_5 = -\frac{54}{30} \\ x_4 + \frac{26}{30}x_5 = \frac{72}{30} \end{cases}$$

En el caso donde $x_5 \in \mathbb{R}$, podemos despejar las variables dependientes:

$$x_1 - 4x_5 = 58 \Rightarrow x_1 = 58 + 4x_5$$

$$x_2 + \frac{148}{30}x_5 = -\frac{2154}{30} \Rightarrow x_2 = -\frac{2154}{30} - \frac{148}{30}x_5 = -\frac{359}{5} - \frac{74}{15}x_5$$

$$x_3 - \frac{2}{30}x_5 = -\frac{54}{30} \Rightarrow x_3 = -\frac{9}{5} + \frac{1}{15}x_5$$

$$x_4 + \frac{26}{30}x_5 = \frac{72}{30} \Rightarrow x_4 = \frac{12}{5} - \frac{13}{15}x_5$$

Por el conjunto solución es dando un $x_5 \in \mathbb{R}$ es:

$$(x_1, x_2, x_3, x_4, x_5) = \left(58 + 4x_5, -\frac{359}{5} - \frac{74}{15}x_5, -\frac{9}{5} + \frac{1}{15}x_5, \frac{12}{5} - \frac{13}{15}x_5, x_5 \right)$$